

Volatility Forecasting: Diagnostic Appendix

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A / Coverage and Reconciliation

This appendix documents checks of retained historical outputs. It is not a fresh model run. All comparative losses use the 4,473-date intersection; gate tables retain each variant's native diagnostic coverage.

Forecast series	Native n	Start	End
GARCH-t	4515	2007-01-05	2024-12-11
Pure LSTM	4515	2007-01-05	2024-12-11
Pure GRU	4515	2007-01-05	2024-12-11
Feature LSTM	4473	2007-03-08	2024-12-11
Feature GRU	4473	2007-03-08	2024-12-11
Residual LSTM	4473	2007-03-08	2024-12-11
Residual GRU	4473	2007-03-08	2024-12-11

What was checked

All seven series have unique prediction dates, finite targets and forecasts, nonnegative targets, and strictly positive stored forecasts. Native-sample MSE and QLIKE reproduce the saved metric file to numerical rounding. Positive stored forecasts do not rule out clipping failures: residual forecasts include values exactly at $1e-12$.

On common dates, the largest absolute difference among stored targets is $5.2e-10$. Targets pass a relative tolerance of $1e-6$ with absolute tolerance $1e-12$, consistent with floating-point rounding. The review retains each file's stored target rather than silently replacing it.

Input scope: yfinance ^GSPC, 2003-2024 including warmup and training. The historical evaluation period does not begin in 2003. Prediction timestamps, corporate data revisions, and full experiment reproducibility still require a fresh end-to-end audit.

B / Historical Loss Comparison



Figure A1. Historical common-date MSE for all seven specifications. Bars start at zero; values are scaled by 10^{-7} . The model order is by family, not a validated ranking.

Saved forecast	MSE / 10 ⁻⁷	QLIKE	Floor hits
GARCH-t	3.058102	-8.085829	0
Pure LSTM	3.566812	-5.272290	0
Pure GRU	3.561952	-3.776601	0
Feature LSTM	3.602508	-4.088043	0
Feature GRU	3.572742	-2.506646	0
Residual LSTM	2.860449	858,837.12	99
Residual GRU	3.082436	4,000,312.63	297

All rows: $n = 4,473$, March 8, 2007 to December 11, 2024. Lower is better for both losses. QLIKE is shown in a table rather than compressing negative and million-scale positive values onto a misleading linear comparison plot.

Interpretation limit: matching dates fixes unequal coverage in the table, but does not correct stale GARCH forecasts, transform inconsistencies, or missing uncertainty estimates.

C / GARCH State-Update Audit

The baseline repeatedly calls forecast(horizon=1) on the same fitted result between parameter refits. No new return is supplied to that result. The returned one-step forecast remains anchored at the fitted endpoint rather than advancing one day at a time.

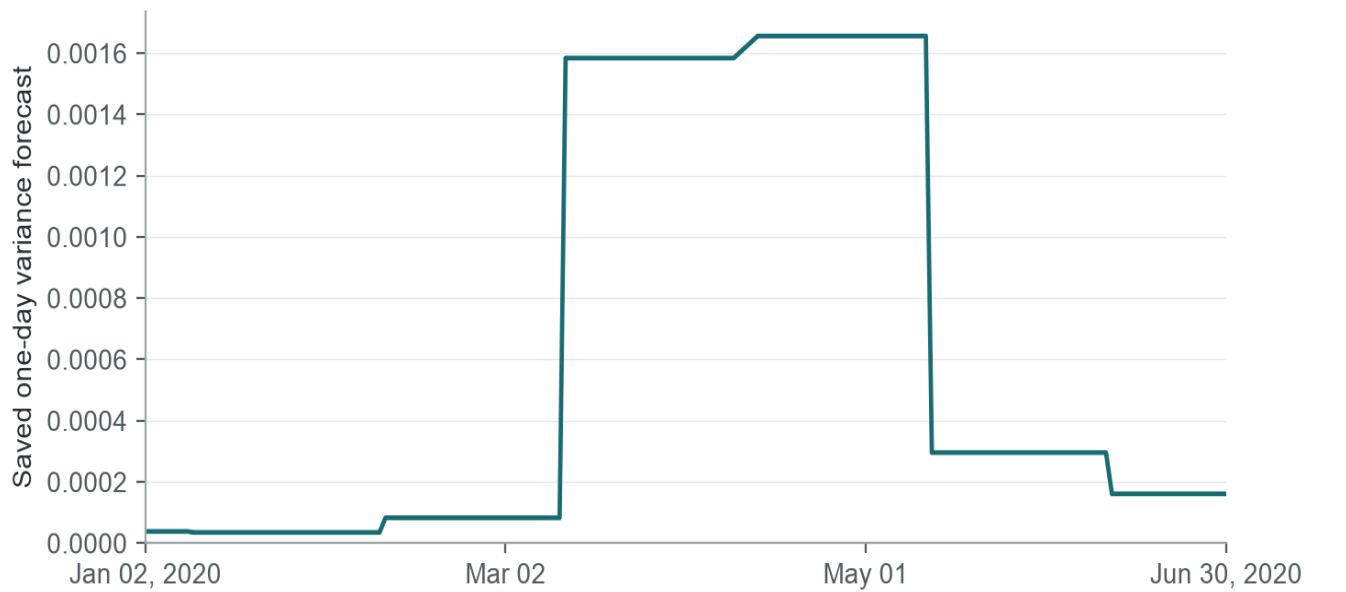


Figure A2. A six-month slice of saved baseline predictions. This plot is selected to show repeated forecasts, not to estimate comparative performance.

Whole-series check	Observed result
Identical to the previous forecast	4,300 of 4,514 transitions (95.26%)
Median / maximum constant run	21 / 21 trading observations
Focused control-flow check	With fitting mocked, changing a return between refits leaves the tested next forecast unchanged.

Why parameter refitting and state updating differ

GARCH parameters may be held fixed for several days while the conditional variance state is updated after each newly observed return. Re-estimation every 21 days is not itself the defect. Reusing the endpoint forecast without the intervening state recursion is the defect. The mock test isolates this control flow; it does not validate a fitted GARCH model or provide a corrected forecast.

Affected artifacts: GARCH baseline; GARCH-feature inputs; residual targets and reconstruction. These dependencies must be regenerated after correction.

D / Numerical and Objective Checks

Floor sensitivity

Prediction floor	Residual LSTM QLIKE	Residual GRU QLIKE
1e-12	858,837.115884	4,000,312.632582
1e-10	8,580.029944	39,997.212392
1e-08	77.559990	394.360908
1e-06	-7.384414	-4.224426

Prediction-only floor sensitivity on 4,473 common dates. The observed-target floor remains 1e-12 for every row. A larger floor mechanically reduces severe underprediction penalties; choosing a floor after inspecting test losses would introduce selection bias.

Target transformation

Implemented forward target map:

$$z = [\log(\max(y, \text{eps})) - \text{train_mean}] / \text{train_std}$$

Implemented inverse:

$$h = \exp(z * \text{train_std} + \text{train_mean}) - \text{eps}$$

$$\text{eps} = 1\text{e-}8$$

Input y	Round-trip output (approximately)
1e-8	0 (small floating-point error before clipping)
1e-6	9.90e-7
1e-4	9.999e-5

A clipping transform is not invertible below its floor. Above the floor, however, subtracting eps introduces an avoidable discrepancy. Any correction must explicitly define behavior at the floor and test the transformed training objective against the forecast estimand.

Checkpoint diagnostics

The existing train/validation gap uses separately minimized losses, potentially from different epochs. A revised log should evaluate both samples at the selected checkpoint. Losses across log-target and signed-residual objectives should not be compared as if they shared a common scale.

E / Gate Diagnostics: Full Summary

Same-date VIX close versus unit-averaged, lag-1-to-20 activations. All 21 Pearson correlations reconcile to the saved summary; Spearman values are independently computed from ranks. Native coverage differs across variants.

Model	Activation	n	Pearson	Spearman
Pure LSTM	candidate	4515	+0.018	+0.043
Pure LSTM	forget	4515	-0.596	-0.221
Pure LSTM	input	4515	+0.167	+0.075
Pure LSTM	output	4515	-0.145	+0.000
Feature LSTM	candidate	4473	-0.098	+0.041
Feature LSTM	forget	4473	-0.322	-0.170
Feature LSTM	input	4473	+0.082	-0.006
Feature LSTM	output	4473	-0.265	-0.098
Residual LSTM	candidate	4473	-0.127	-0.012
Residual LSTM	forget	4473	-0.104	+0.231
Residual LSTM	input	4473	+0.362	+0.277
Residual LSTM	output	4473	+0.139	+0.324
Pure GRU	candidate	4515	+0.023	+0.074
Pure GRU	reset	4515	+0.227	+0.179
Pure GRU	update	4515	+0.114	+0.277
Feature GRU	candidate	4473	+0.059	+0.105
Feature GRU	reset	4473	+0.062	-0.062
Feature GRU	update	4473	+0.088	+0.267
Residual GRU	candidate	4473	+0.024	+0.104
Residual GRU	reset	4473	+0.270	+0.232
Residual GRU	update	4473	-0.026	+0.078

Candidate activations are included with the gate summaries for consistency with the saved files; they are not sigmoid gates. These are pooled correlations across rolling refits, without uncertainty estimates, adjustment for serial dependence, or a causal interpretation.

F / Verification and Provenance

Category	Observed defects / evidence	Assessment
Forecast calculation	Stale GARCH state; target inverse mismatch.	Not ready for validated model comparison.
Evaluation	Seven native metric rows reconcile; all seven aligned to 4,473 dates. Residual floor failures persist.	Arithmetic checked; model validity unresolved.
Interpretation	Original gate-mechanism and model-superiority claims exceed the available evidence.	Reframed as hypotheses and historical diagnostics.

Category	Observed defects / evidence	Assessment
Report design	Original narrative mixed unequal samples and omitted key implementation limitations.	Replaced with explicit dates, definitions, limitations, and source-backed figures.
Completeness	No fresh training, repeated seeds, formal loss-difference tests, or economic backtest.	Not presented as completed validation.
Reproducibility	Public code and aggregate evidence; local prediction CSVs are not bundled.	Artifact hashes recorded; full independent rerun pending.

Source trail

Aggregate values, sensitivity outputs, focused test results, eight source-file SHA-256 hashes, and four reviewed code-blob hashes are published in `reports/review_metrics.json`. The review inspected the report narrative and relevant code; the 54-page legacy diagnostic deck was used as context, not certified page by page.

Open limitations

No fresh yfinance download; no retraining; no line-by-line audit of every repository module; no confirmation that all historical charts used the exact same artifact version. The new tables and figures use only the explicitly checked sources. The legacy report-generation script still describes the historical workflow.

Research provenance

The underlying experiments originated in a collaborative research project. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers have been omitted; the byline does not imply sole authorship of the original experiments.